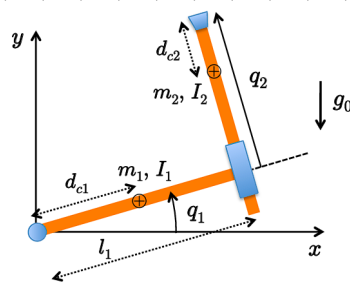


Exercise 1

The dynamic model of the planar RP robot in Fig. 1 moving in a vertical plane can be written in the usual form as

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau. \quad (1)$$

- Define two different matrices $S_1(q, \dot{q})$ and $S_2(q, \dot{q})$ that factorize the Coriolis and centrifugal terms (i.e., yielding $S_i(q, \dot{q})\dot{q} = c(q, \dot{q})$, for $i = 1, 2$), such that $\dot{M} - 2S_1$ is skew-symmetric, while $\dot{M} - 2S_2$ is not.
- Give the explicit symbolic expressions of the terms appearing in the definition of the momentum-based residual vector $r \in \mathbb{R}^2$ that allows detection and isolation of collisions.
- Are there situations in which collision forces $F_K \in \mathbb{R}^2$ in the plane of motion lead to poor or no detection, or to incorrect isolation of the involved link? Discuss the issue.



First of all i need to write the dynamic model:

$$P_{c1} = \begin{cases} d_{c1} \dot{c}_1 \\ d_{c1} s_1 \end{cases} \Rightarrow v_{c1} = \begin{cases} -d_{c1} s_1 \dot{q}_1 \\ -d_{c1} c_1 \dot{q}_1 \end{cases} \Rightarrow T_1 = \frac{1}{2} m_1 d_{c1}^2 \dot{q}_1^2 + \frac{1}{2} I_1 \dot{q}_1^2 = \frac{1}{2} \dot{q}_1^2 (I_1 + m_1 d_{c1}^2)$$

$$P_2 = \begin{cases} l_1 c_1 + (q_2 - d_{c2}) s_1 \\ l_1 s_1 - (q_2 - d_{c2}) c_1 \end{cases} \Rightarrow v_{c2} = \begin{cases} -l_1 s_1 \dot{q}_1 + \dot{q}_2 s_1 + (q_2 - d_{c2}) c_1 \dot{q}_1 \\ l_1 c_1 \dot{q}_1 - \dot{q}_2 c_1 + (q_2 - d_{c2}) s_1 \dot{q}_1 \end{cases}$$

$$v_{c2}^T v_{c2} = l_1^2 \dot{q}_1^2 + \dot{q}_2^2 + q_2^2 \dot{q}_1^2 - d_{c2}^2 \dot{q}_1^2 - 2l_1 \dot{q}_1 \dot{q}_2 - 2q_2 d_{c2} \dot{q}_1^2$$

$$T_2 = \frac{1}{2} m_2 (l_1^2 \dot{q}_1^2 + \dot{q}_2^2 + q_2^2 \dot{q}_1^2 - d_{c2}^2 \dot{q}_1^2 - 2l_1 \dot{q}_1 \dot{q}_2 - 2q_2 d_{c2} \dot{q}_1^2) + \frac{1}{2} I_2 \dot{q}_2^2$$

$$= \frac{1}{2} m_2 (\dot{q}_2^2 - 2l_1 \dot{q}_1 \dot{q}_2) + \frac{1}{2} \dot{q}_1^2 (I_2 + m_2 [l_1^2 + q_2^2 - d_{c2}^2 - 2q_2 d_{c2}])$$

$$M_{11} = I_1 + m_1 d_{c1}^2 + I_2 + m_2 l_1^2 - m_2 d_{c2}^2 + q_2 (m_2 q_2 - 2d_{c2}) = p_1 + q_2 (p_2 q_2 + p_3)$$

$$M_{22} = m_2 = p_2$$

$$M_{12} = -m_2 l_1 = p_4$$

$$M = \begin{pmatrix} p_1 + p_2 q_2^2 + p_3 q_2 & p_2 \\ p_2 & p_4 \end{pmatrix}$$

Now i compute g

$$U = g_0 (m_1 d_{c1} s_1 + m_2 [l_1 s_1 - (q_2 - d_{c2}) c_1])$$

$$\frac{\partial U}{\partial q_1} = g_0 (m_1 d_{c1} c_1 + m_2 [l_1 c_1 + (q_2 - d_{c2}) s_1]) = g_0 [p_5 c_1 - p_4 c_1 + p_2 q_2 s_1 + \frac{1}{2} p_2 p_3 s_1]$$

$$\frac{\partial U}{\partial q_2} = g_0 m_2 c_1 = g_0 p_2 c_1$$

Now i compute the Christoffel symbols

$$C_{111} = 0 \Rightarrow C_{11} = C_{111} \dot{q}_1 + C_{121} q_2 = 0$$

$$C_{112} = p_2 q_2 + \frac{1}{2} p_3 \quad C_{12} = C_{112} \dot{q}_1 + C_{122} \dot{q}_2 = \dot{q}_1 (p_2 q_2 + \frac{1}{2} p_3)$$

$$C_{211} = -p_2 q_2 - \frac{1}{2} p_3 \quad C_{21} = C_{211} \dot{q}_1 + C_{221} \dot{q}_2 = -\dot{q}_1 (p_2 q_2 + \frac{1}{2} p_3)$$

$$C_{212} = 0 \quad C_{22} = C_{212} \dot{q}_1 + C_{222} \dot{q}_2 = 0$$

$$C_{122} = 0$$

$$C_{222} = 0$$

$$S_1 = \dot{q}_1 (p_2 q_2 + \frac{1}{2} p_3) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad S_2 = S_1 + \begin{pmatrix} -\dot{q}_2 & 0 \\ 0 & \dot{q}_1 \end{pmatrix}$$

$$p_5 = m_1 d_{c1} c_1 - p_4 c_1 + p_2 q_2 s_1 + \frac{1}{2} p_2 p_3 s_1$$

p_6

$$\begin{cases} (p_1 + p_2 q_2^2 + p_3 q_2) \ddot{q}_1 + p_2 \ddot{q}_2 + g_0(p_5 c_1 - p_4 c_1 + p_2 q_2 s_1 + \frac{1}{2} p_6 s_1) + \dot{q}_1 \dot{q}_2 (p_2 q_2 + \frac{1}{2} p_3) = \tau_1 \\ p_2 \ddot{q}_1 + p_4 \ddot{q}_2 + g_0 p_2 c_1 - \dot{q}_1^2 (p_2 q_2 + \frac{1}{2} p_3) = \tau_2 \end{cases}$$

$$\begin{pmatrix} \ddot{q}_1 & q_2^2 \ddot{q}_1 + \ddot{q}_2 + g_0 q_2 s_1 + \dot{q}_1 \dot{q}_2 q_2 & q_2 \ddot{q}_1 + \frac{1}{2} \dot{q}_1 \dot{q}_2 & -g_0 c_1 & g_0 c_1 & g_0 \frac{1}{2} s_1 \\ 0 & \ddot{q}_1 + g_0 c_1 - \dot{q}_1^2 q_2 & -\dot{q}_1^2 \frac{1}{2} & \ddot{q}_2 & 0 & 0 \end{pmatrix} p = \tau$$

$\tau \succ$

Now we compute α for the residual method:

$$\alpha_1 = g_1 \quad \alpha_2 = g_2 - \dot{q}_1^2 (2p_2 q_2 + p_3) \quad p = M \dot{q} = \begin{cases} (p_1 + p_2 q_2^2 + p_3 q_2) \dot{q}_1 + p_2 \dot{q}_2 \\ p_2 \dot{q}_1 + p_4 \dot{q}_2 \end{cases}$$

A collision force F_K is not detected if $F_K \in \mathcal{N}(J^T)$. Instead, if F_K is applied on the second link but $(J^T F_K)^T \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$, then we see the collision happening on the first link, happens with zero force orthogonal to the second link.

Exercise 2

Consider an actuated pendulum with a link of (known) length l that moves without friction in the vertical plane. The pendulum is driven by a DC motor at the base and carries a heavy and *unknown* payload. Assume that the motor inertia and the mass and inertia of the link are negligible with respect to the payload, which should be seen as a concentrated mass m at the tip of the pendulum. The drive gain k_i of the current-to-torque relation $\tau = k_i i_m$ of the motor is *unknown*, and only the motor current i_m can be commanded.

Design an adaptive control law for i_m that achieves global asymptotic tracking of a smooth desired trajectory $\theta_d(t)$ for the joint angle θ of the pendulum. Provide a (sketch of) proof of your result.

the position is $p_c = \begin{cases} l \cos \theta \\ l \sin \theta \end{cases} \Rightarrow v_c = \begin{cases} -l \sin \theta \dot{\theta} \\ l \cos \theta \dot{\theta} \end{cases} \Rightarrow T = \frac{1}{2} m l^2 \dot{\theta}^2 \Rightarrow M = m l^2$

$$U = m g_0 l \sin \theta \Rightarrow \gamma = m g_0 l \cos \theta$$

$$\Rightarrow m l^2 \ddot{\theta} + m g_0 l \cos \theta = \kappa_i z_m, \quad z_m = \text{control input.}$$

We consider the dynamic coefficients

$$\frac{1}{\kappa_i} m l^2 \ddot{\theta} + \frac{1}{\kappa_i} m g_0 l \cos \theta = z_m \Rightarrow e_1 \ddot{\theta} + e_2 \cos \theta = z_m = \begin{pmatrix} \ddot{\theta} & \cos \theta \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}$$

We have to learn p_1, p_2 . We define $\dot{\theta}_r = \dot{\theta}_d + \gamma (\theta_d - \theta)$ and

We choose z_m as follows

$$\sigma = \dot{\theta}_r - \dot{\theta}$$

$$e = \theta_d - \theta$$

$$z_m = \begin{pmatrix} \ddot{\theta}_r & \cos \theta \end{pmatrix} \begin{pmatrix} \hat{p}_1 \\ \hat{p}_2 \end{pmatrix} + \kappa_p e + \kappa_d \dot{e}$$

and
$$\begin{cases} \dot{\hat{e}}_1 = \lambda_1 \ddot{\theta}_r \cdot \sigma \\ \dot{\hat{e}}_2 = \lambda_2 c\theta \cdot \sigma \end{cases}$$

In this way, when $\hat{p} \rightarrow p$, we will have $e = -\frac{\kappa_d}{\kappa_p} \dot{e}$

Exercise 3

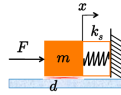
With reference to Fig. 2, a mass $m > 0$ moves under the action of a control force F , in the presence of viscous friction with coefficient $d > 0$, and interacts with a stiff environment. We would like to regulate the contact force F_c to a constant desired value $F_d > 0$. The contact force is measured by a load cell of stiffness $k_s > 0$, i.e., $F_c = k_s x$ where $x = 0$ corresponds to the initial contact position with the environment. Consider a class of control laws of the form

$$F = \alpha k_f (F_d - F_c) + \beta F_d, \quad (2)$$

where $k_f > 0$, and with:

1. $(\alpha, \beta) = (0, 1)$ [pure feedforward];
2. $(\alpha, \beta) = (1, 0)$ [pure proportional feedback];
3. $(\alpha, \beta) = (1, 1)$ [combined feedback/feedforward].

- For each of the three above control cases, check the system equilibrium and verify its stability properties, giving a proof of your statements (e.g., via Lyapunov/LaSalle, or using Laplace analysis in view of the linearity of the system) and briefly discussing the benefits and limitations of each law.
- How robust are these results with respect to uncertainty in the knowledge of the physical parameters m , d , and k_s ?
- Explain what happens under the action of the above control laws when there is no environment present ($F_c \equiv 0$). Would the system reach some form of steady state?



$$m\ddot{x} = F - \kappa_s x - d\dot{x} \quad \kappa_s x = F_c \Rightarrow F_d = \kappa_s x_d \Rightarrow x_d = \frac{F_d}{\kappa_s}$$

case 1)

$$m\ddot{x} = \beta \kappa_s x_d - \kappa_s x - d\dot{x} \Rightarrow$$

Laplace: $s^2 m x - \beta \kappa_s x_d + \kappa_s x + d s x = 0$

$$x (m s^2 + d s + \kappa_s) = \beta \kappa_s x_d$$

$$x = \frac{\beta \kappa_s x_d}{(m s^2 + d s + \kappa_s)} \quad \text{poles in:} \quad s_{1,2} = \frac{-d \pm \sqrt{d^2 - 4 m \kappa_s}}{2m}$$

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} x(s) = x_d \Rightarrow \lim_{t \rightarrow \infty} F_c = F_d \Rightarrow \text{desired force achieved}$$

For any m, d, κ_s positive.

$$\text{case 2) } m\ddot{x} = \kappa_s (F_d - F_c) - \kappa_s x - d\dot{x} \Rightarrow$$

$$m\ddot{x} = \kappa_s \kappa_s x_d - \kappa_s x (1 + \kappa_s) - d\dot{x}$$

$$m x s^2 + \kappa_s (1 + \kappa_s) x + d x s = \kappa_s \kappa_s x_d$$

$$\frac{x}{x_d} = \frac{\kappa_s \kappa_s}{(m s^2 + d s + \kappa_s (1 + \kappa_s))} \quad \text{stable for any } \kappa_s > 0$$

$$\lim_{s \rightarrow 0} \frac{x}{x_d} = \frac{\kappa_s}{1 + \kappa_s} \neq 1 \Rightarrow \text{there will be a steady state error.}$$